

BANARAS HINDU UNIVERSITY

4pt]Department of Geology – – – Faculty of Science 4pt] Practical Record : Identification of Rock – Forming Minerals in Hand Specimen (30 Specimens) 6pt] _____

#	Mineral	Form	Color	Streak	Luster	Cleavage	Fracture	Hard.	S.G.	System	Formula	Diagn
1	Quartz	Prismatic	Colorless	White	Vitreous	No cleavage	Conchoidal	7	2.65	Hexagonal	SiO ₂	Hardne
Orthoclase	Tabular	Flesh red,	White	Pearly	Two set at °	Uneven	6	2.56	Monoclinic	KAlSi ₃ O ₈	Luster, Hardness, Cleavage 3	Microcl
	White	White	Pearly	One set	Conchoidal	6	2.56	Triclinic	KAlSi ₃ O ₈	Cleavage, Luster, Hardness 4	Albite	Prisma
	White	Pearly	Two set	Uneven	6	2.62	Triclinic	NaAlSi ₃ O ₈	Cleavage, Hardness 5	Labradorite	Massive	Blue - C
Indistinct	Vitreous	Two set at °	Uneven	6–6.5	2.70	Triclinic	(Ca, Na)(Al, Si) ₄	Hardness, Luster, Cleavage 6	Muscovite	Lamellar Flakes	Transp.–Silver	White
Pearly	One set	Conch.– Uneven	2.5	2.80	Monoclinic	KAl ₂ (AlSi ₃ O ₁₀)(OH) ₂	Sheet form, Luster, Transp. 7	Biotite	Lamellar Flakes	Dark brown–black	White–Grey	Pearly
One set	Conch.– Uneven	2.5–3	2.7–3.4	Monoclinic	K(Mg, Fe) ₃ (Al	Cleavage, Luster 8	Nepheline	Massive	White, Grey, Blue	Colorless	Greasy–Waxy	Poor
Uneven	5.5–6	2.60	Hexagonal	Na ₃ KAl ₄ Si ₄ O ₁₆	Greasy Luster 9	Tremolite	Prismatic	White–Pale green	White	Silky	Two perfect	Uneven
5.5–6	3.0–3.3	Monoclinic	Ca ₂ Mg ₅ Si ₈ O ₂₂ (C	Luster, Cleavage 10	Augite	Prismatic	Black	White	Vitreous	Two perfect	Uneven	5–6
3.2–3.4	Monoclinic	(Ca, Mg, Fe, Na)(Mg, Fe, Al) ₂ (Si, Al) ₂ O ₆	Blende	Massive	Dark green	White	Vitreous	Two set	Uneven	5–6	3.25	
Monoclinic	(K, Na) _{0–1} (Ca, N	Luster, Cleavage 12	Olivine	Massive	Olive green	White	Dull	Indistinct	Undetermi	6.7	3.20	Orthorh
(Mg, Fe) ₃ Si ₄ O ₁₃	Talc	Massive	White	White	Dull–Pearly	Perfect (Basal)	Indistinct	1	2.80	Monoclinic	Mg ₃ Si ₄	Mg ₃ Si ₄
Hard Pearl	Serpentine	Massive	Light green	White	Greasy	Absent	Uneven	4	2.60	Monoclinic	Mg ₆ Si ₄ O ₁₀ (OH) ₈	Luster
Apatite	Massive	Dark green	White	Vitreous	One set	Conchoidal	5	3.20	Hexagonal	Ca ₅ (PO ₄) ₃ (O	Fracture, Cleavage 16	Calcite
Tabular	White	White	Vitreous	Three set perfect	Uneven	3	2.70	Hexagonal	CaCO ₃	Cleavage, Hardness 17	Dolomite	Massive

#	Mineral	Form	Color	Streak	Luster	Cleavage	Fracture	Hard.	S.G.	System	Formula	Diagn
Grey	White	Dull	3 set Rhombo.	Conchoidal	3.5–4	2.85	Hexagonal	CaMg(CO ₃)	Cleavage, Luster 18	Kaolinite	Massive	Grey
White	Dull	Indistinct	Uneven	2–2.5	2.60	Triclinic	Al ₂ Si ₂ O ₅ (OH) ₄	Luster, Hardness 19	Garnet	Dodecahedron	Reddish Brown	Undete
Vitreous	Absent	Uneven	7–8	3.50	Isometric	A ₃ B ₂ (SiO ₄) ₃	Crystal system, Luster 20	Kyanite	Bladed	Blue to white	White	Pearly
Two set perfect	Uneven	6	3.60	Triclinic	Al ₂ SiO ₅	Form, Luster, Cleavage 21	Sillimanite	Acicular	White–Olive green	Undetermined	Vitreous	Indistin
Uneven	6.5	3.20	Orthorhombic	Al ₂ SiO ₅	Form, Hardness, Luster 22	Andalusite	Prismatic	Brown to red	Undetermi	Dull	Indistinct	Subcon
7.5	3.18	Orthorhombic	Al ₂ SiO ₅	Hardness, Form, Luster 23	Staurolite	Prismatic	Dark brown	White	Vitreous	Absent	Subconchoidal	7–7.5
3.75	Monoclinic	Fe ₂ Al ₉ Si ₄ O ₂₃ (	Hardness, Luster 24	Topaz	Prismatic	Light pink–yellow	Undetermined	Vitreous	One perfect	Subconchoidal	8	3.5–3.6
Orthorhombic	Al ₂ SiO ₄ (OH, F) ₂	Hardness, Cleavage 25	Tourmaline	Prismatic	Black	White	Resinous	Poor	Subconchoidal	7–7.5	2.90	Hexago
(Na, +	Hardness, Cleavage 26	Corundum	Granular	Pink	Indistinct	Adamantine	Absent	Uneven	9	4.00	Hexagonal	Al ₂ O ₃
Hardness 27	Gypsum	Tabular	Colorless	White	Vitreous	1 set	Conchoidal	2	2.32	Monoclinic	CaSO ₄ · 2H ₂ O	Cleavag Hardne
Beryl	Massive	Green	Undetermined	Vitreous	Poor	Subconchoidal	7.5–8	2.80	Hexagonal	Be ₃ Al ₂ (SiO ₃) ₆	Hardness, Cleavage 29	Agate
Massive	Grey	Undetermined	Waxy	Absent	Uneven	6.5–7	2.65	Trigonal	SiO ₂	Luster, Cleavage 30	Magnesite	Massive
Pale yellow–brown	White	Porcelainous	3 set	Conchoidal	3.5–5	3.00	Hexagonal	MgCO ₃	Luster, Fracture			